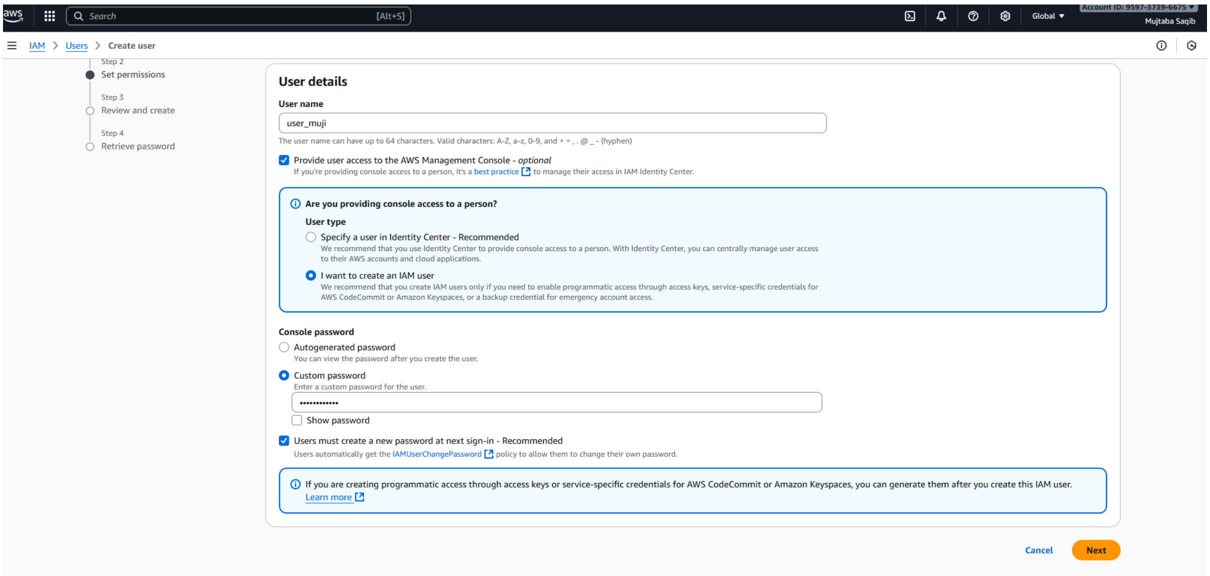
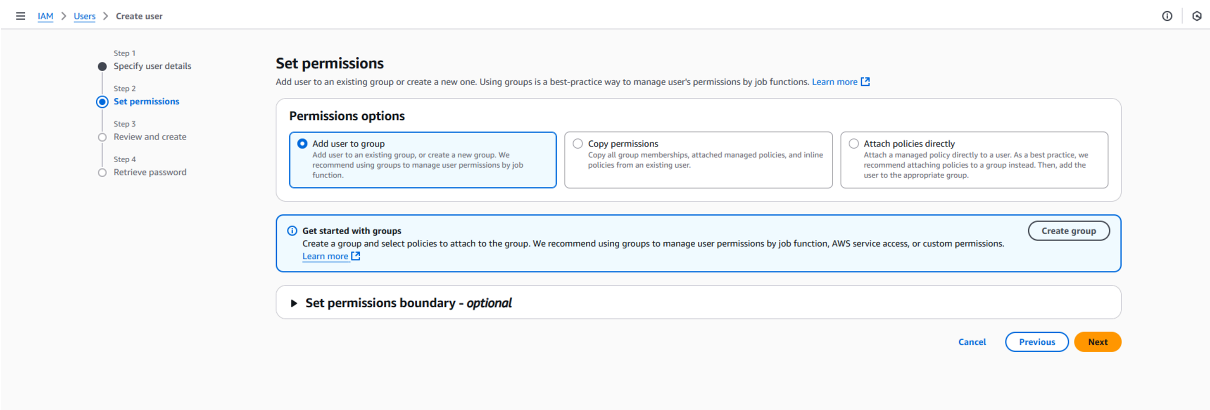
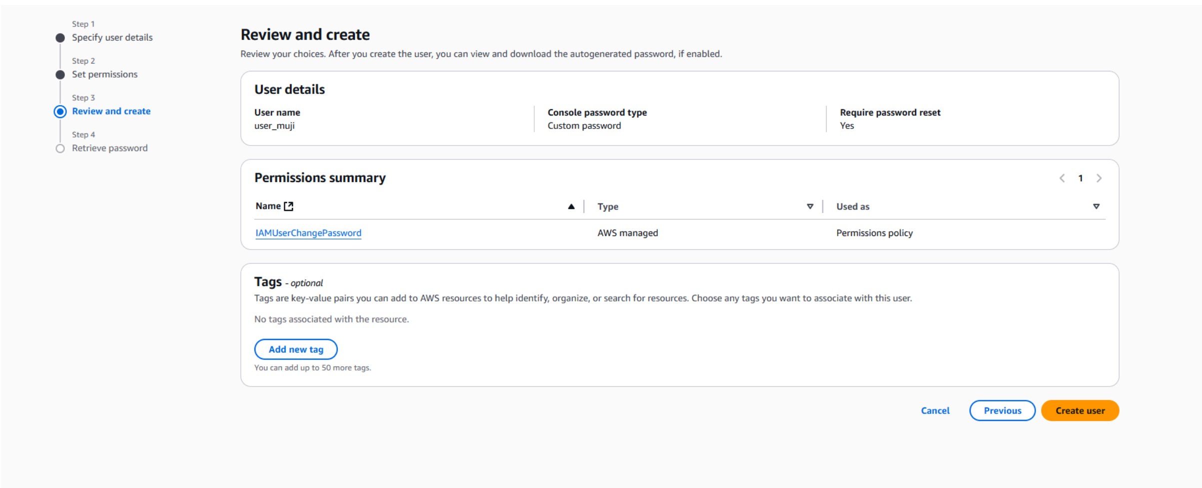
AWS Lab 01:



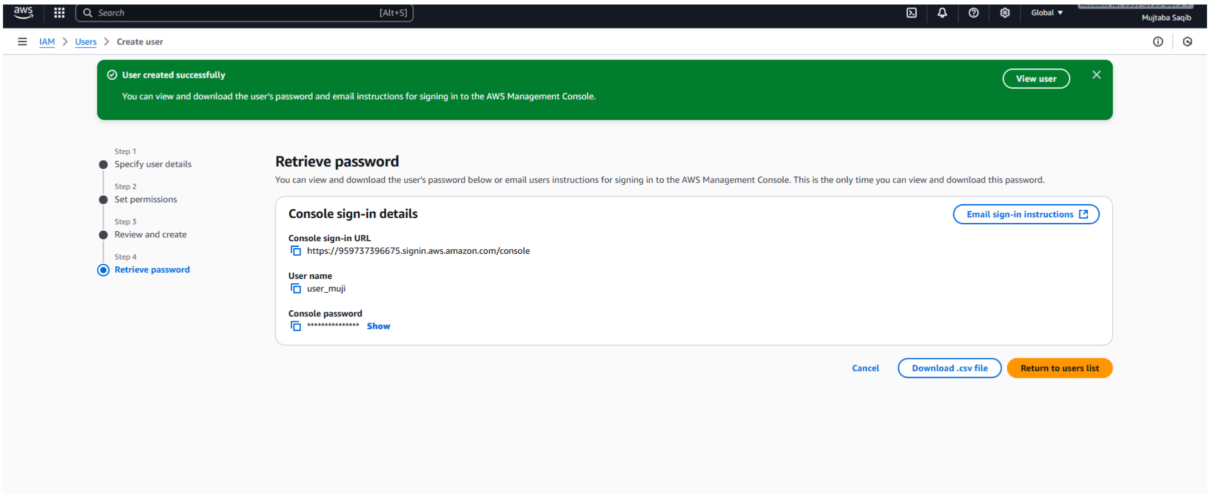
Search [Alt+S]
Global
Mujtaba Saqib
IAM > Users > Create user
Step 2
Set permissions
Step 3
Review and create
Step 4
Retrieve password
User details
User name
user_muji
The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)
Provide user access to the AWS Management Console - optional
If you're providing console access to a person, it's a best practice to manage their access in IAM Identity Center.
Are you providing console access to a person?
User type
Specify a user in Identity Center - Recommended
We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.
I want to create an IAM user
We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.
Console password
Autogenerated password
You can view the password after you create the user.
Custom password
Enter a custom password for the user.
Show password
Users must create a new password at next sign-in - Recommended
Users automatically get the IAMUserChangePassword policy to allow them to change their own password.
If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user.
Learn more
Cancel
Next

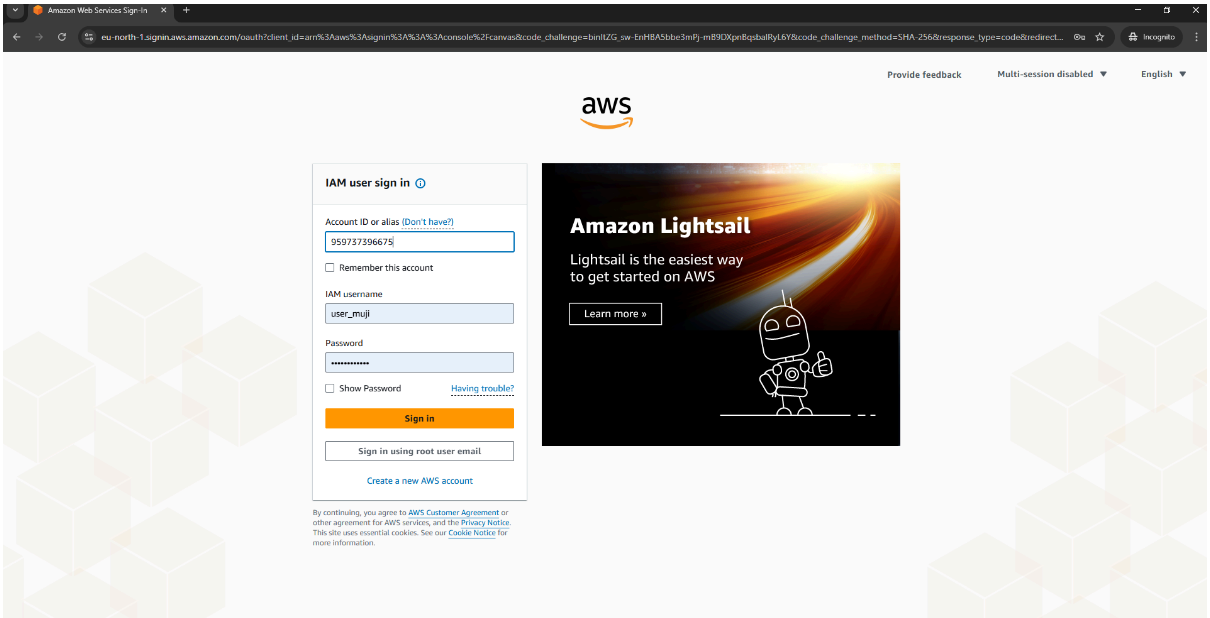


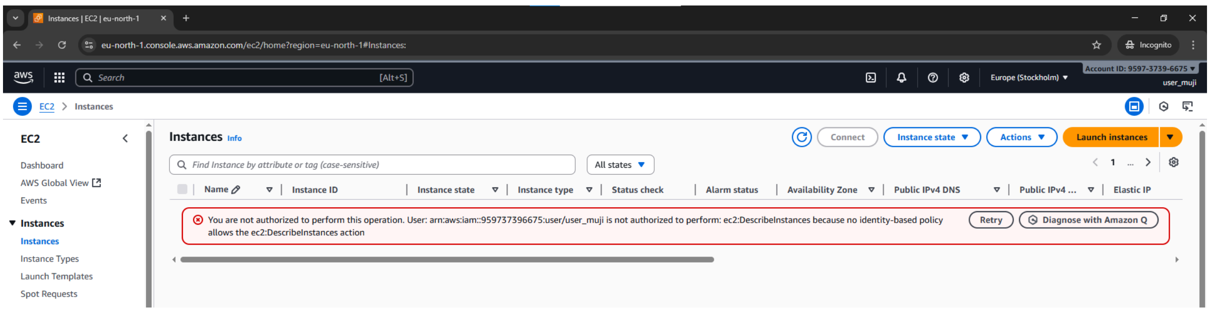
IAM > Users > Create user
Step 1
Specify user details
Step 2
Set permissions
Step 3
Review and create
Step 4
Retrieve password
Set permissions
Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. Learn more
Permissions options
Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.
Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.
Attach policies directly
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.
Get started with groups
Create a group and select policies to attach to the group. We recommend using groups to manage user permissions by job function, AWS service access, or custom permissions.
Learn more
Create group
Set permissions boundary - optional
Cancel
Previous
Next



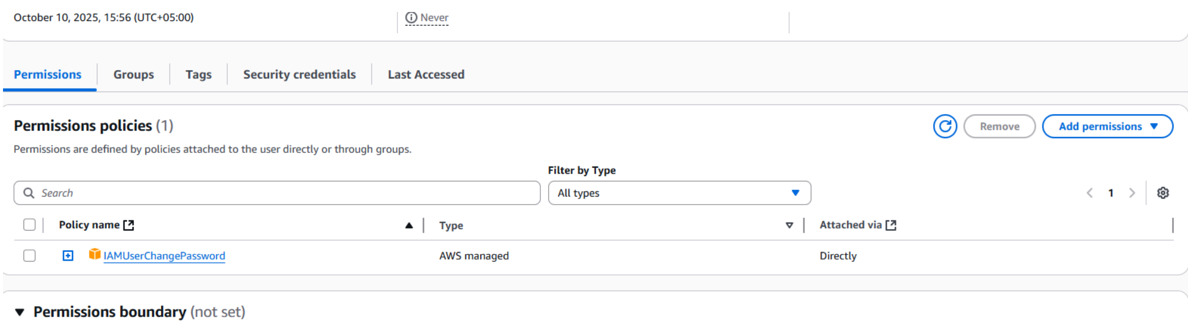
Step 1
Specify user details
Step 2
Set permissions
Step 3
Review and create
Step 4
Retrieve password
Review and create
Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.
User details
User name
user_muji
Console password type
Custom password
Require password reset
Yes
Permissions summary
Name
Type
Used as
IAMUserChangePassword
AWS managed
Permissions policy
Tags - optional
Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.
No tags associated with the resource.
Add new tag
You can add up to 50 more tags.
Cancel
Previous
Create user

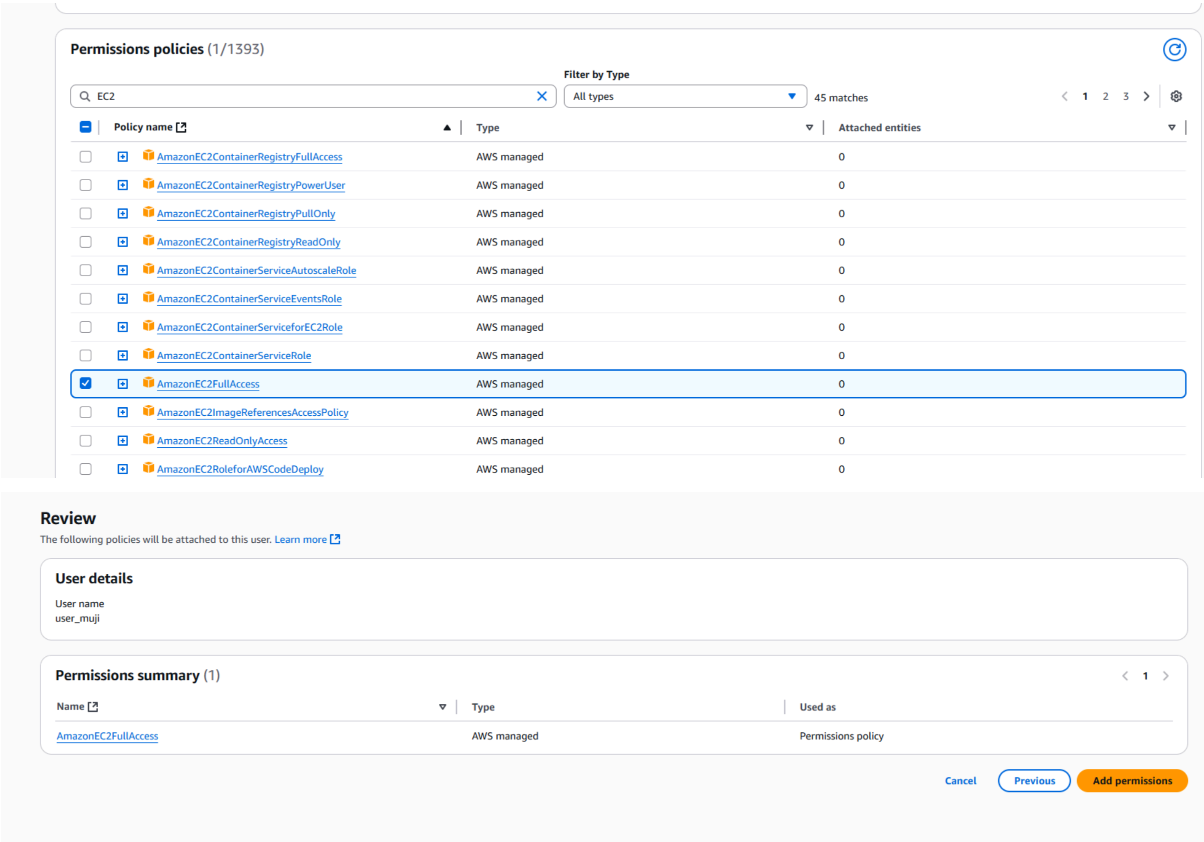




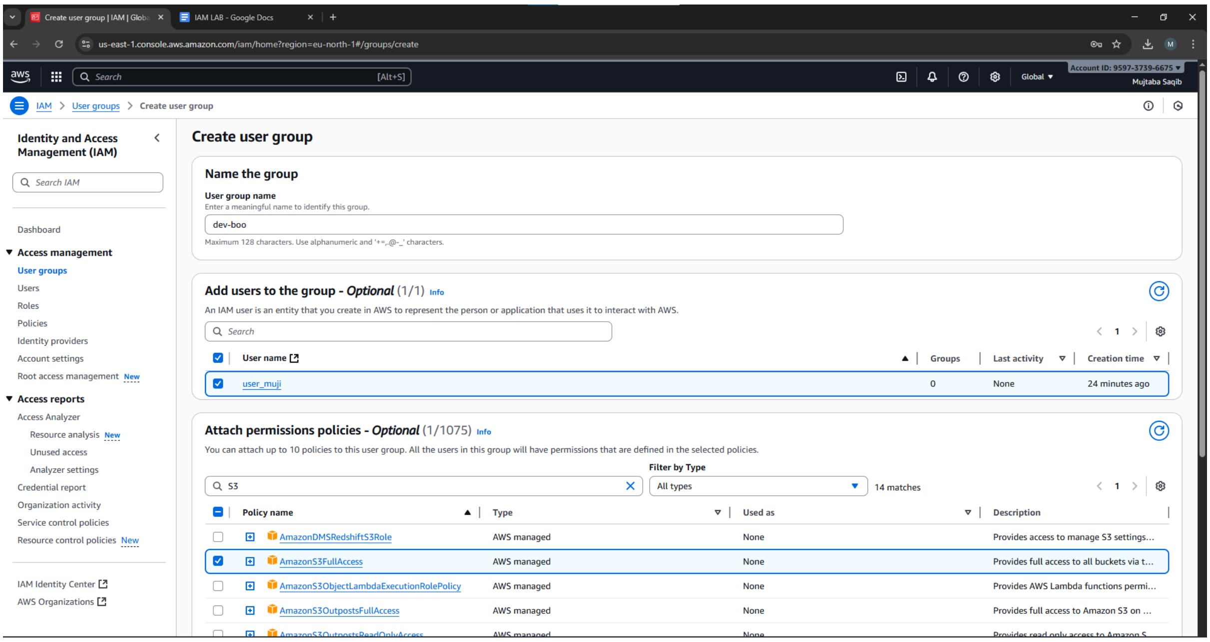


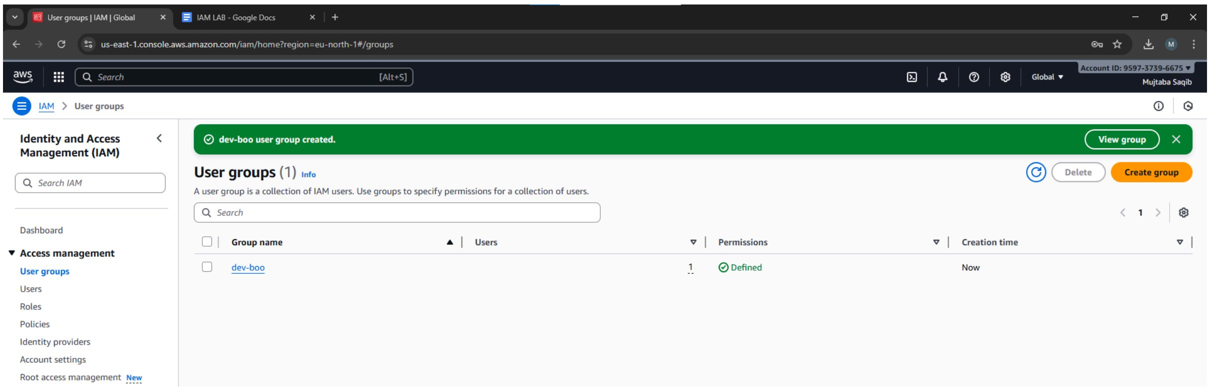
Direct method



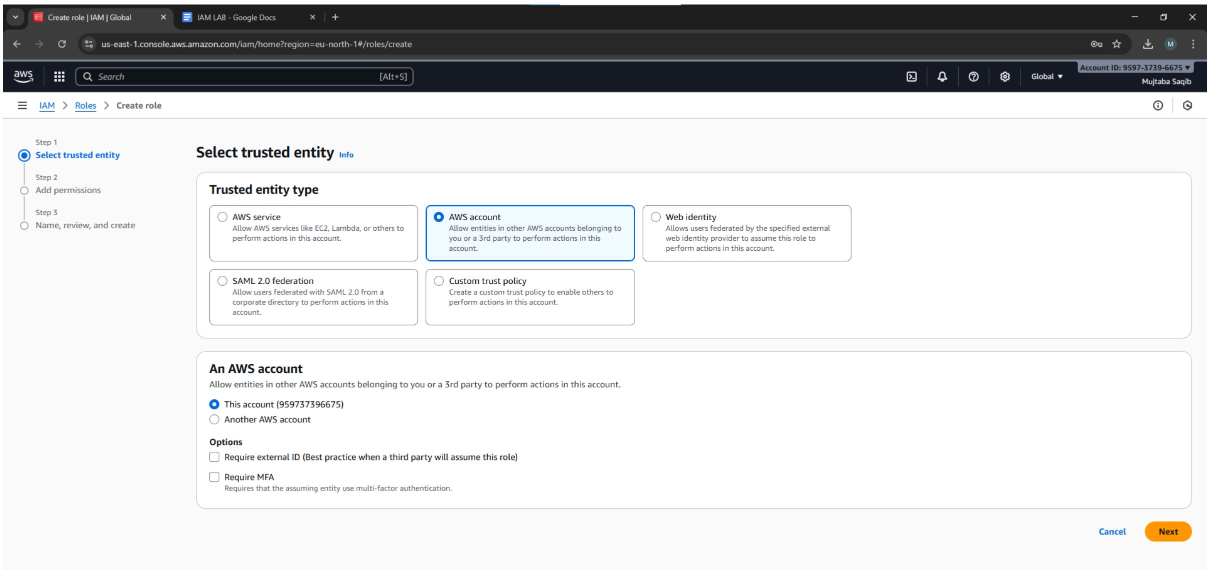


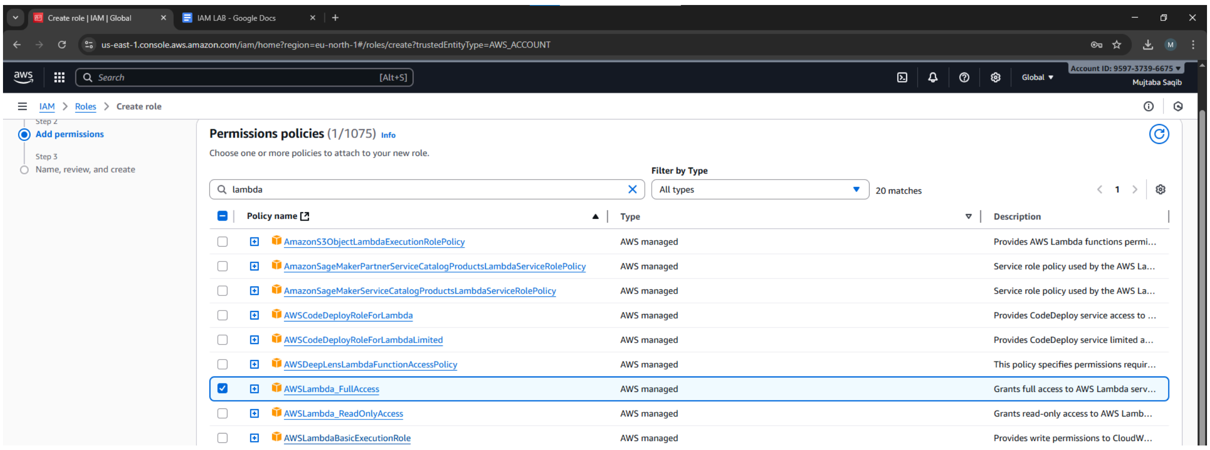
Create group (3-4 people)

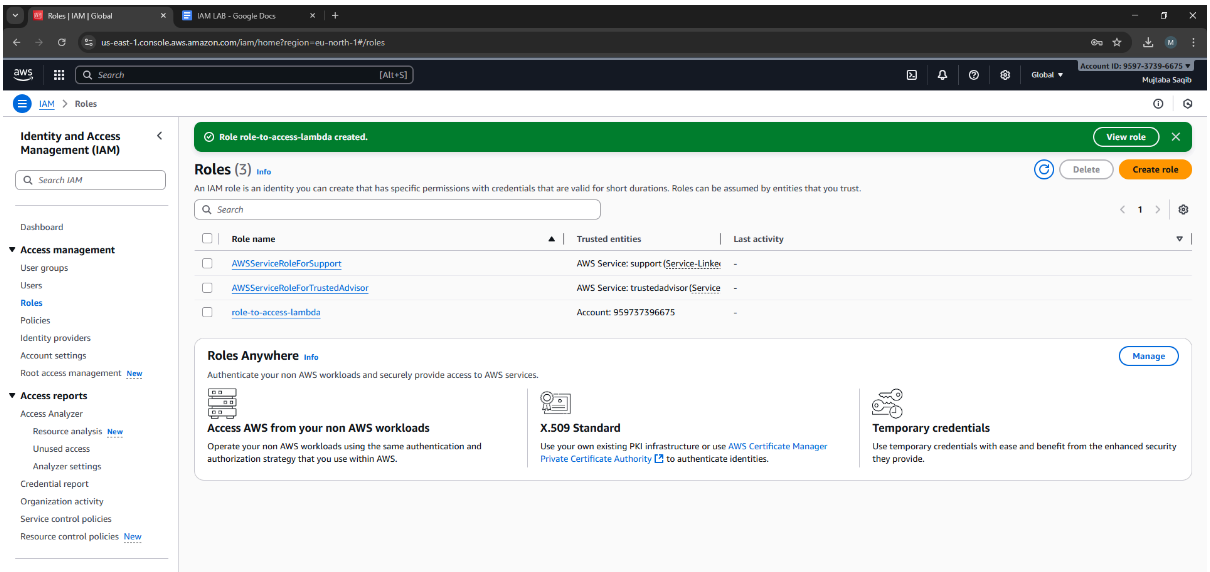


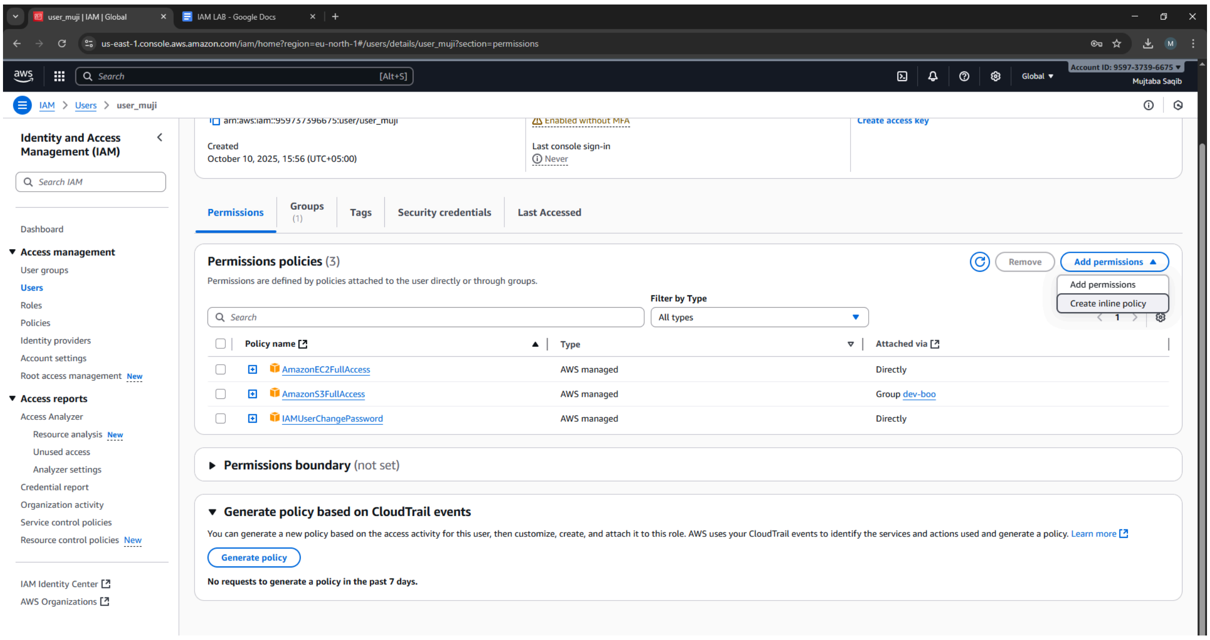

User groups | IAM | Global
IAM LAB - Google Docs
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/groups
Account ID: 9597-3739-6675
Mujtaba Saqib
IAM > User groups
Identity and Access Management (IAM)
Search IAM
Dashboard
Access management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management New
dev-boo user group created.
View group
User groups (1) Info
Delete
Create group
A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.
Group name
Users
Permissions
Creation time
dev-boo
1
Defined
Now

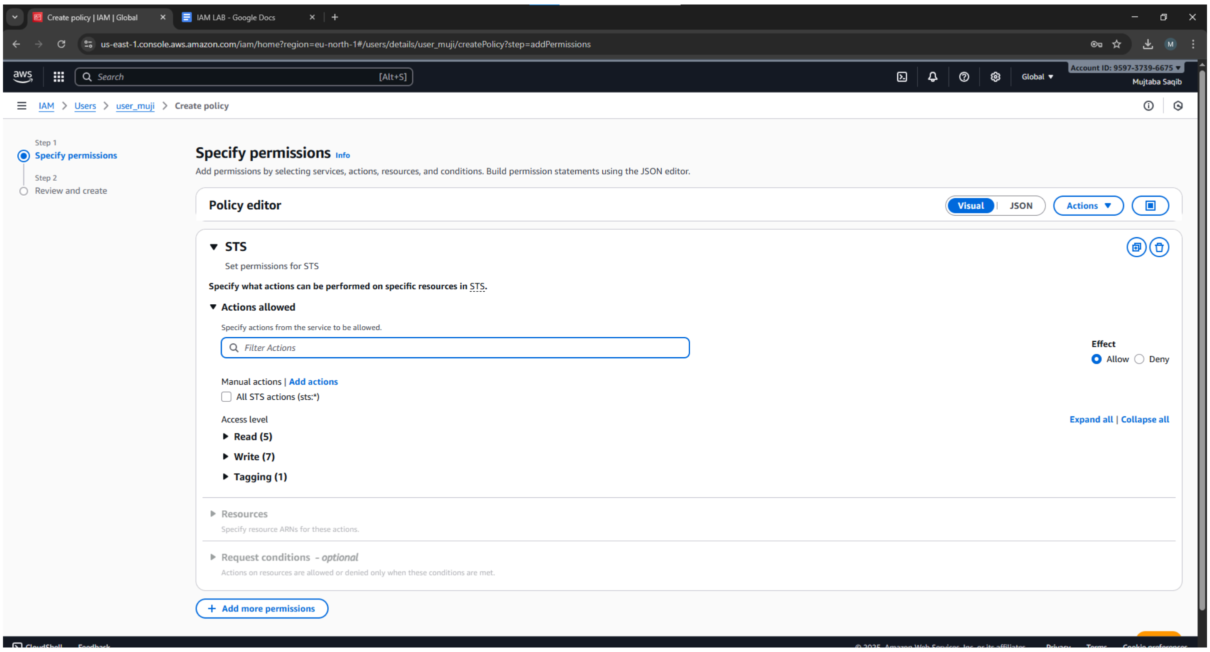
Create role

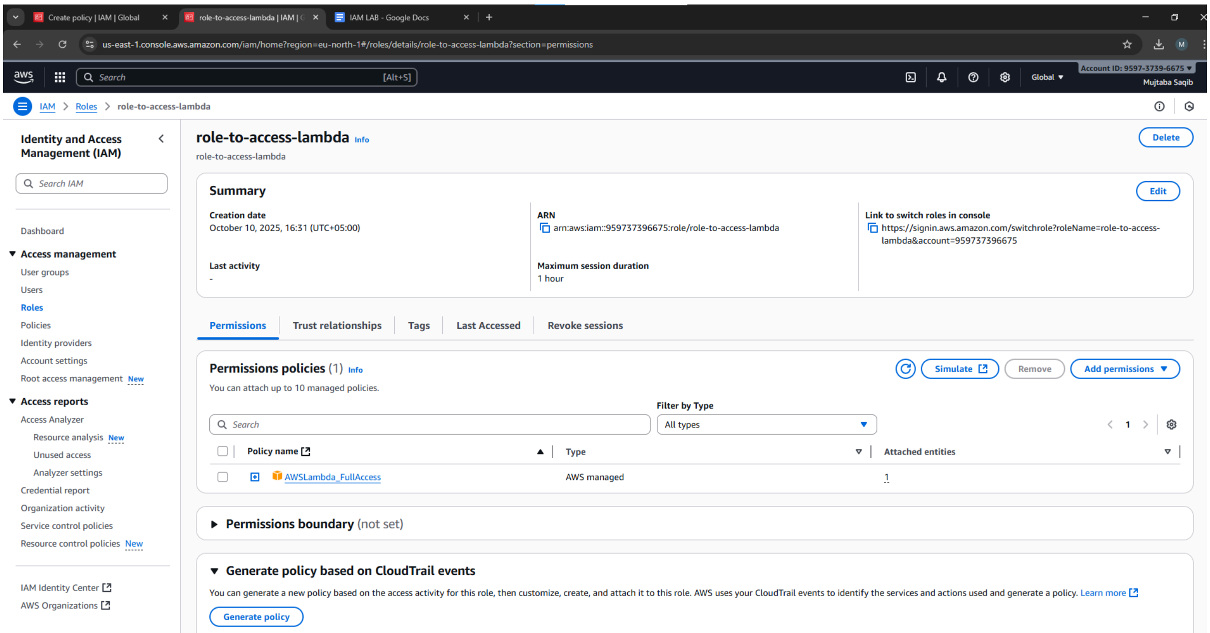

Create role | IAM | Global
IAM LAB - Google Docs
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles/create
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Select trusted entity Info
Trusted entity type
AWS service
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
Web identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.
An AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
This account (959737396675)
Another AWS account
Options
Require external ID (Best practice when a third party will assume this role)
Require MFA
Requires that the assuming entity use multi-factor authentication.
Cancel
Next


Create role | IAM | Global
IAM LAB - Google Docs
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles/create?trustedEntityType=AWS_ACCOUNT
IAM > Roles > Create role
Add permissions
Step 3
Name, review, and create
Permissions policies (1/1075) Info
Choose one or more policies to attach to your new role.
Filter by Type
lambda
All types
20 matches
Policy name
Type
Description
AmazonS3ObjectLambdaExecutionRolePolicy
AWS managed
Provides AWS Lambda functions permi...
AmazonSageMakerPartnerServiceCatalogProductsLambdaServiceRolePolicy
AWS managed
Service role policy used by the AWS La...
AmazonSageMakerServiceCatalogProductsLambdaServiceRolePolicy
AWS managed
Service role policy used by the AWS La...
AWSCodeDeployRoleForLambda
AWS managed
Provides CodeDeploy service access to ...
AWSCodeDeployRoleForLambdaLimited
AWS managed
Provides CodeDeploy service limited a...
AWSDeepLensLambdaFunctionAccessPolicy
AWS managed
This policy specifies permissions requir...
AWSLambda_FullAccess
AWS managed
Grants full access to AWS Lambda serv...
AWSLambda_ReadOnlyAccess
AWS managed
Grants read-only access to AWS Lamb...
AWSLambdaBasicExecutionRole
AWS managed
Provides write permissions to CloudW...

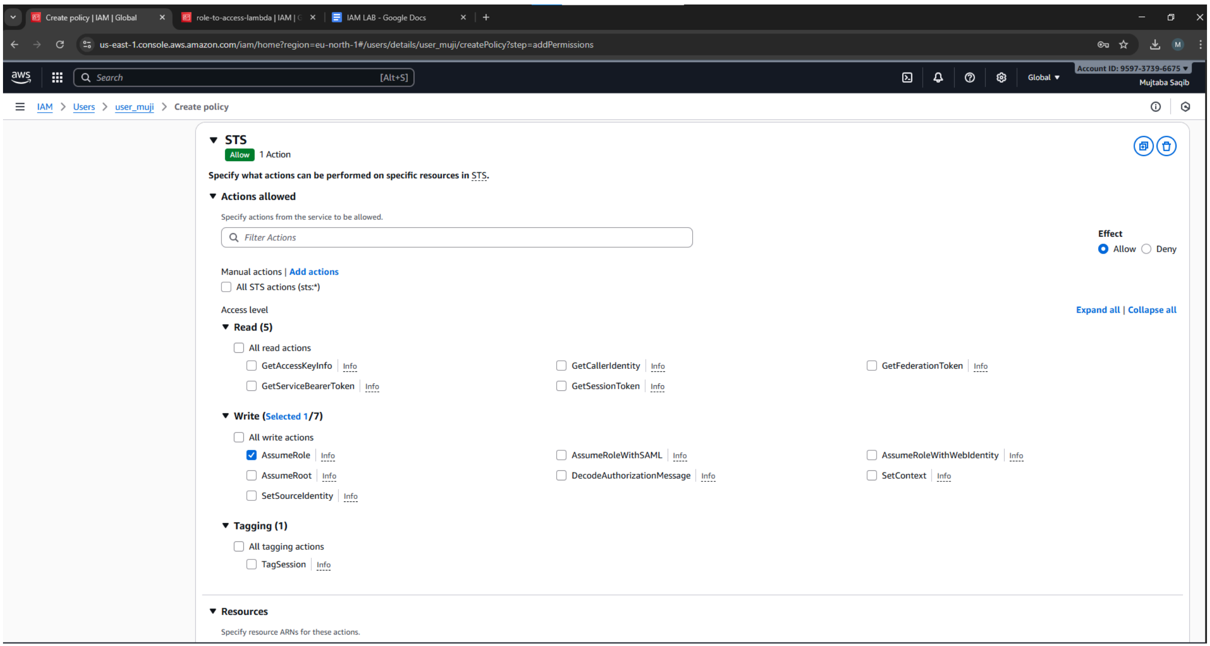

Roles | IAM | Global
IAM LAB - Google Docs
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/roles
Search
[Alt+S]
Global
Account ID: 9597-3739-6675
Mujtaba Saqib
IAM
Roles
Identity and Access Management (IAM)
Search IAM
Dashboard
Access management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management New
Access reports
Access Analyzer
Resource analysis New
Unused access
Analyzer settings
Credential report
Organization activity
Service control policies
Resource control policies New
Role role-to-access-lambda created.
View role
Roles (3) Info
Delete
Create role
An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.
Search
Role name
Trusted entities
Last activity
AWSServiceRoleForSupport
AWS Service: support (Service-Linked
-
AWSServiceRoleForTrustedAdvisor
AWS Service: trustedadvisor (Service
-
role-to-access-lambda
Account: 959737396675
-
Roles Anywhere Info
Manage
Authenticate your non AWS workloads and securely provide access to AWS services.
Access AWS from your non AWS workloads
Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.
X.509 Standard
Use your own existing PKI infrastructure or use AWS Certificate Manager Private Certificate Authority to authenticate identities.
Temporary credentials
Use temporary credentials with ease and benefit from the enhanced security they provide.


user_muji | IAM | Global
IAM LAB - Google Docs
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/users/details/user_muji?section=permissions
Search
[Alt+S]
Global
Account ID: 9597-3739-6675
Mujtaba Saqib
IAM
Users
user_muji
Created
October 10, 2025, 15:56 (UTC+05:00)
Last console sign-in
Never
Create access key
Identity and Access Management (IAM)
Search IAM
Dashboard
Access management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management New
Access reports
Access Analyzer
Resource analysis New
Unused access
Analyzer settings
Credential report
Organization activity
Service control policies
Resource control policies New
IAM Identity Center
AWS Organizations
Permissions
Groups (1)
Tags
Security credentials
Last Accessed
Permissions policies (3)
Remove
Add permissions
Permissions are defined by policies attached to the user directly or through groups.
Add permissions
Create inline policy
Filter by Type
Search
All types
Policy name
Type
Attached via
AmazonEC2FullAccess
AWS managed
Directly
AmazonS3FullAccess
AWS managed
Group dev-boo
IAMUserChangePassword
AWS managed
Directly
Permissions boundary (not set)
Generate policy based on CloudTrail events
You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. Learn more
Generate policy
No requests to generate a policy in the past 7 days.

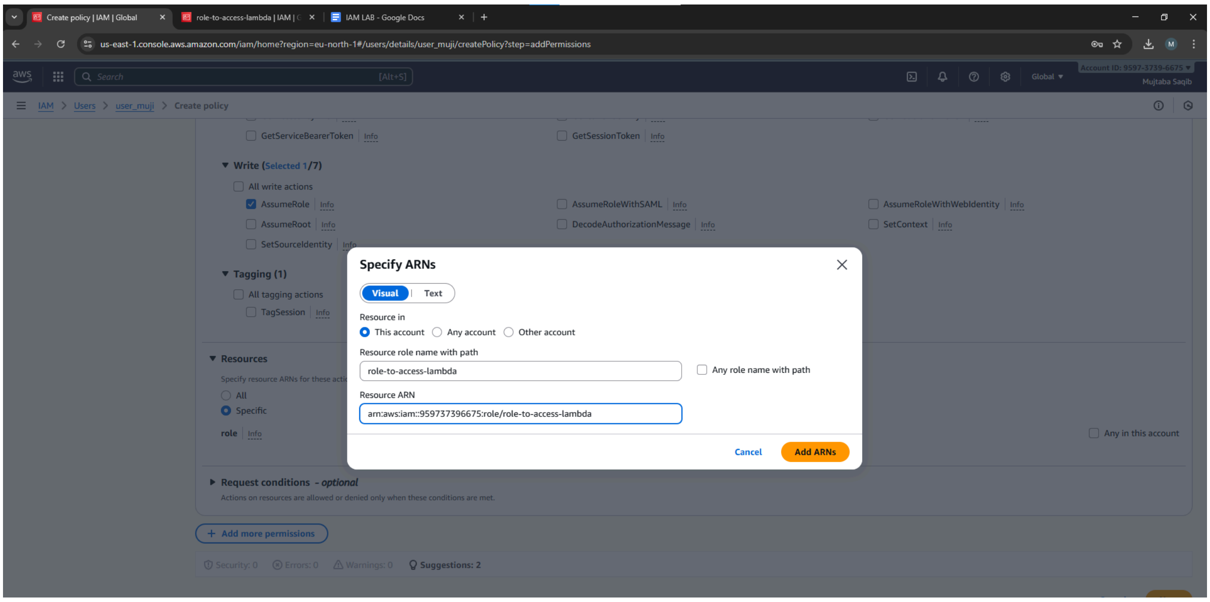


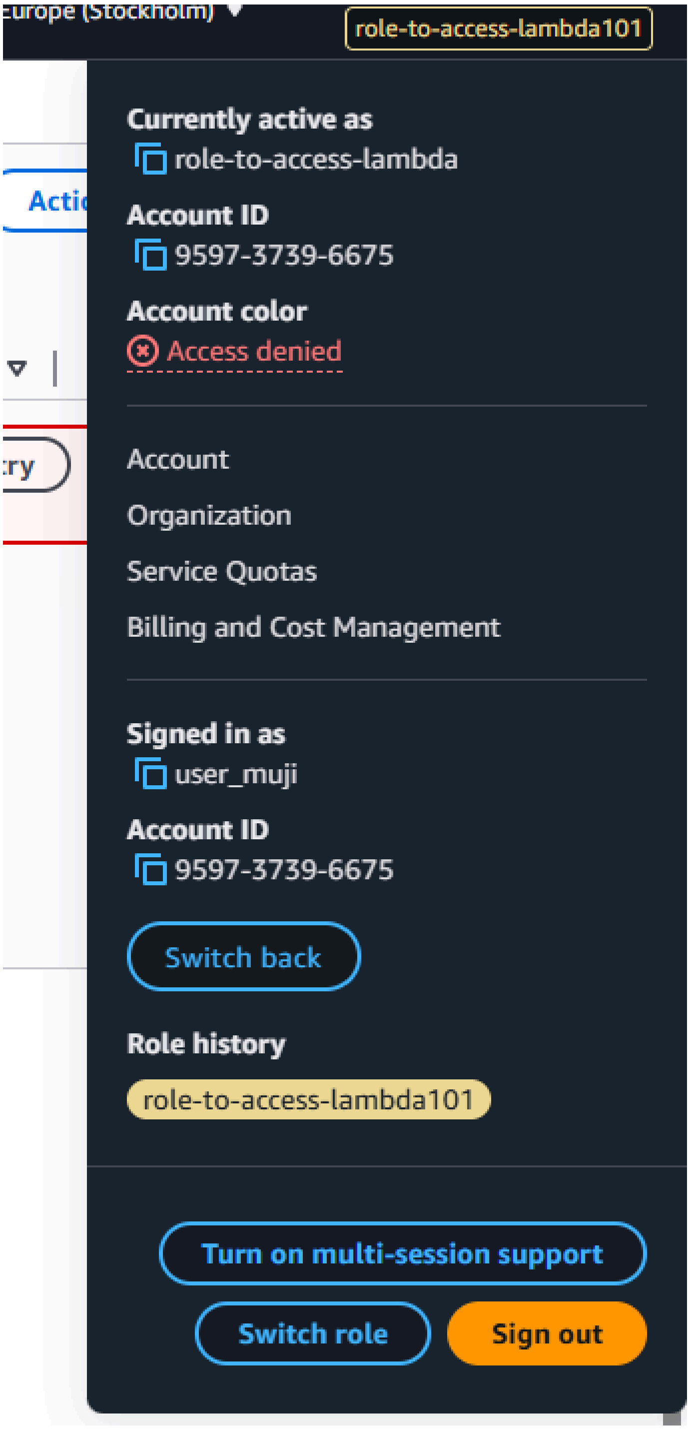


Copy role arn number

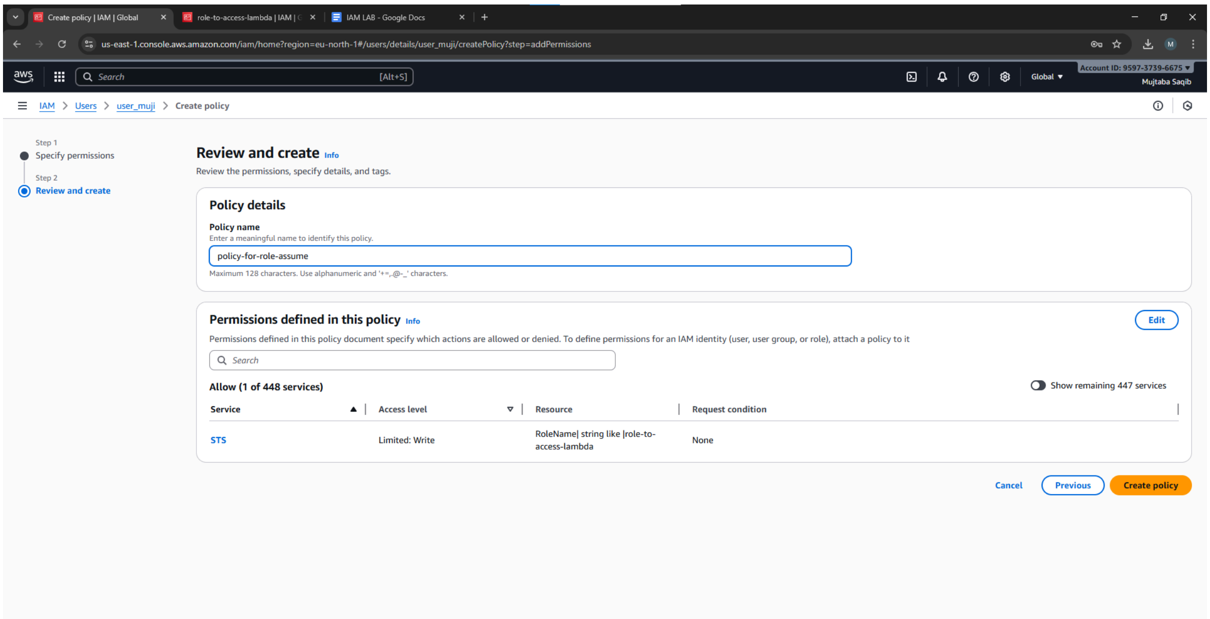


Add arns

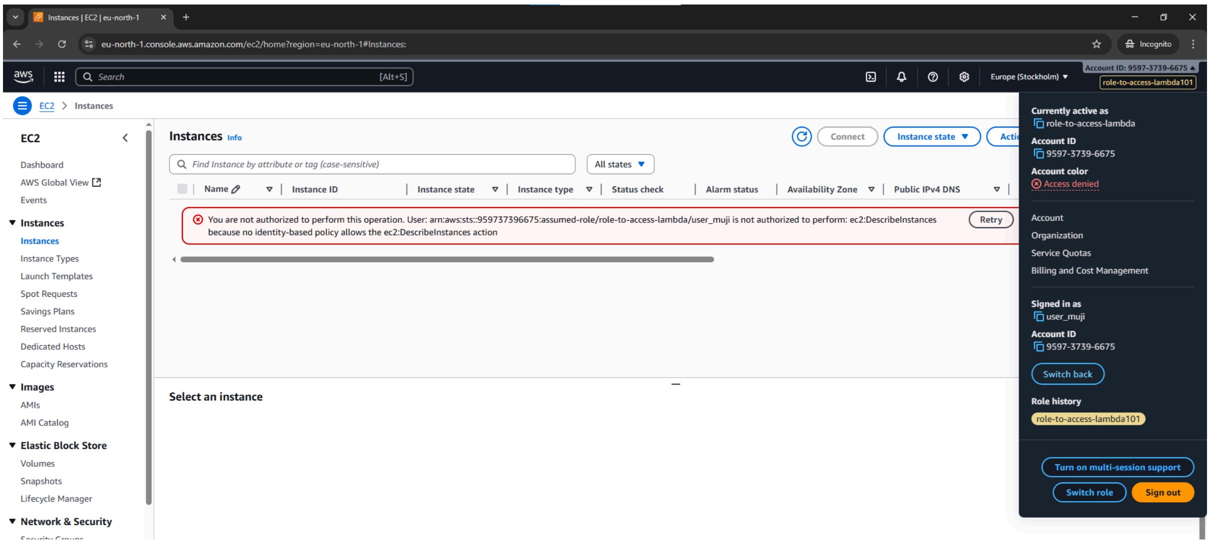



Europe (Stockholm)
role-to-access-lambda101
Currently active as
role-to-access-lambda
Account ID
9597-3739-6675
Account color
Access denied
Account
Organization
Service Quotas
Billing and Cost Management
Signed in as
user_muji
Account ID
9597-3739-6675
Switch back
Role history
role-to-access-lambda101
Turn on multi-session support
Switch role
Sign out

Switch role



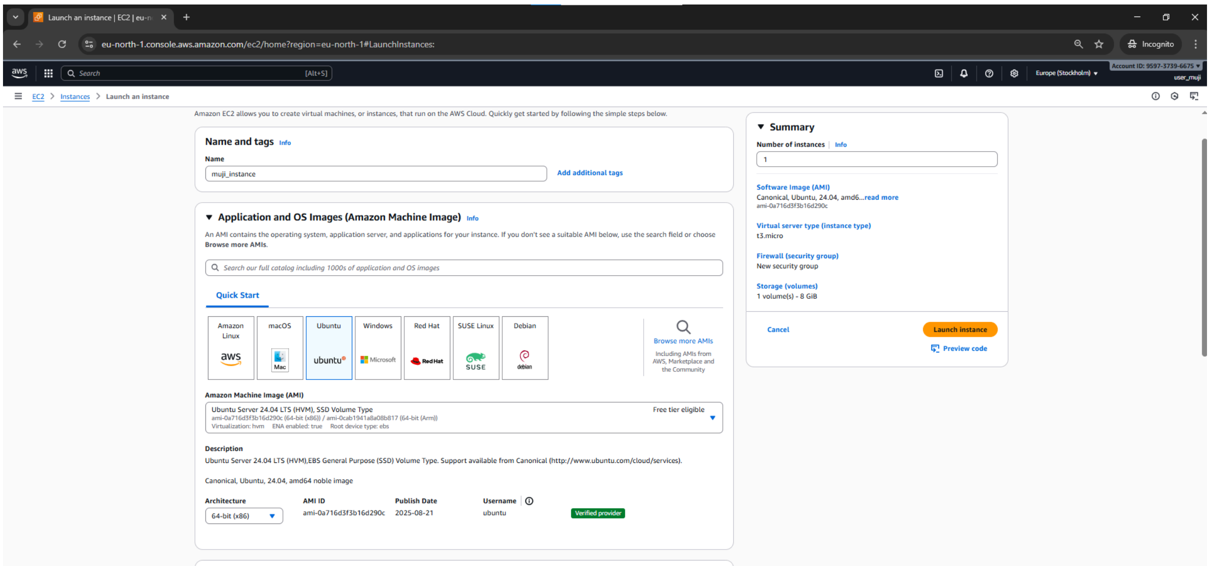
Create policy | IAM | Global
role-to-access-lambda | IAM |
IAM LAB - Google Docs
us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/users/details/user_muji/createPolicy?step=addPermissions
Account ID: 9597-3739-6675
Mujtaba Saqib
IAM > Users > user_muji > Create policy
Step 1
Specify permissions
Step 2
Review and create
Review and create
Review the permissions, specify details, and tags.
Policy details
Policy name
Enter a meaningful name to identify this policy.
policy-for-role-assume
Permissions defined in this policy
Edit
Allow (1 of 448 services)
Show remaining 447 services
Service
Access level
Resource
Request condition
STS
Limited: Write
RoleName| string like |role-to-access-lambda
None
Cancel
Previous
Create policy

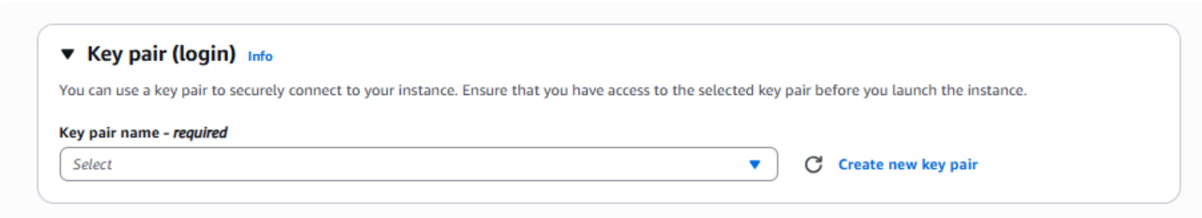


Instances | EC2 | eu-north-1
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances:
Europe (Stockholm)
Account ID: 9597-3739-6675
role-to-access-lambda101
EC2 > Instances
Instances
Connect
Instance state
All states
You are not authorized to perform this operation. User: arn:aws:sts::959737396675:assumed-role/role-to-access-lambda/user_muji is not authorized to perform: ec2:DescribeInstances because no identity-based policy allows the ec2:DescribeInstances action
Retry
Select an instance
Currently active as
role-to-access-lambda
Account ID
9597-3739-6675
Account color
Access denied
Account
Organization
Service Quotas
Billing and Cost Management
Signed in as
user_muji
Account ID
9597-3739-6675
Switch back
Role history
role-to-access-lambda101
Turn on multi-session support
Switch role
Sign out

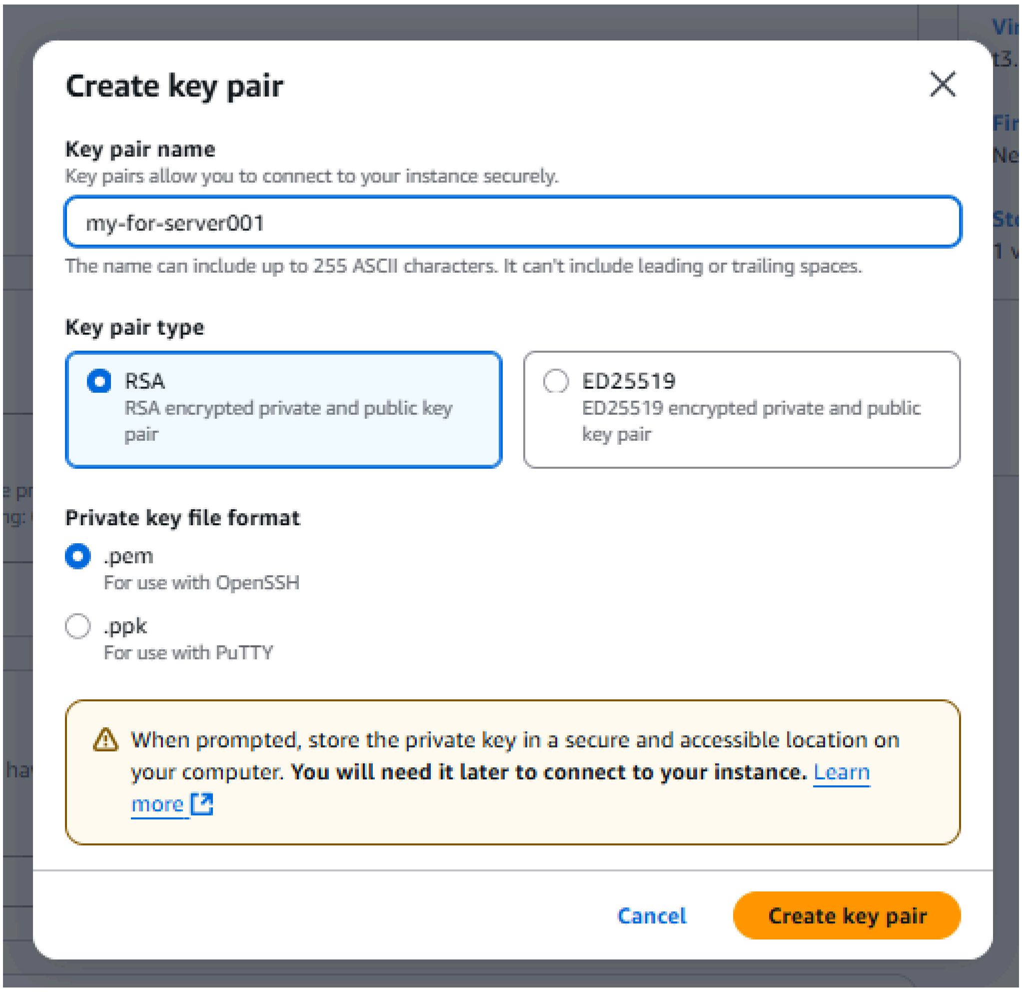
Switch back

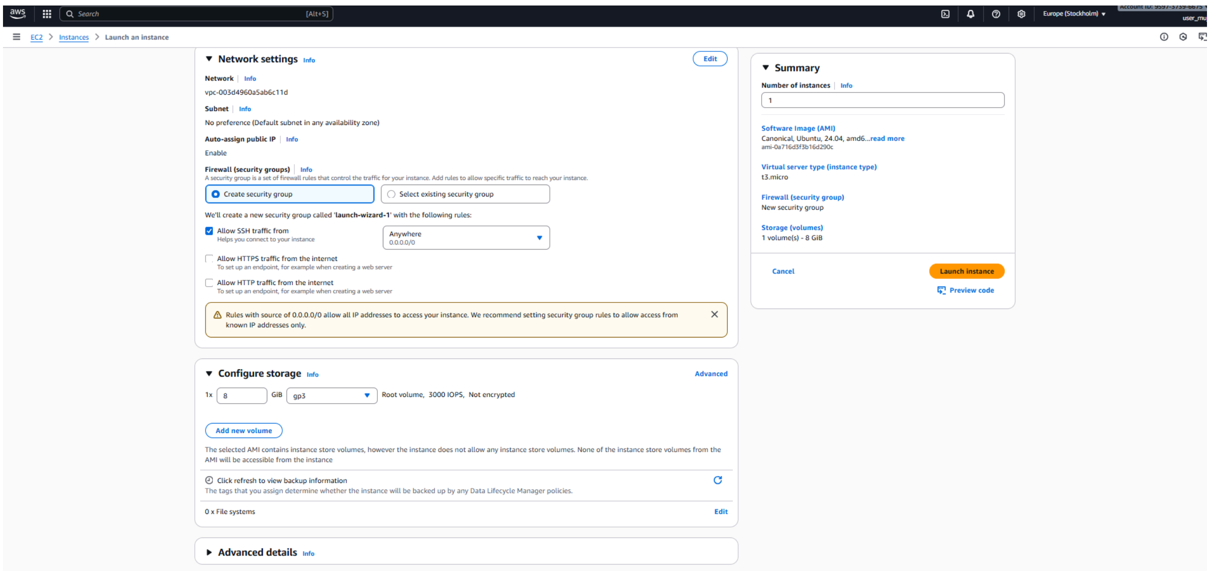
EC2

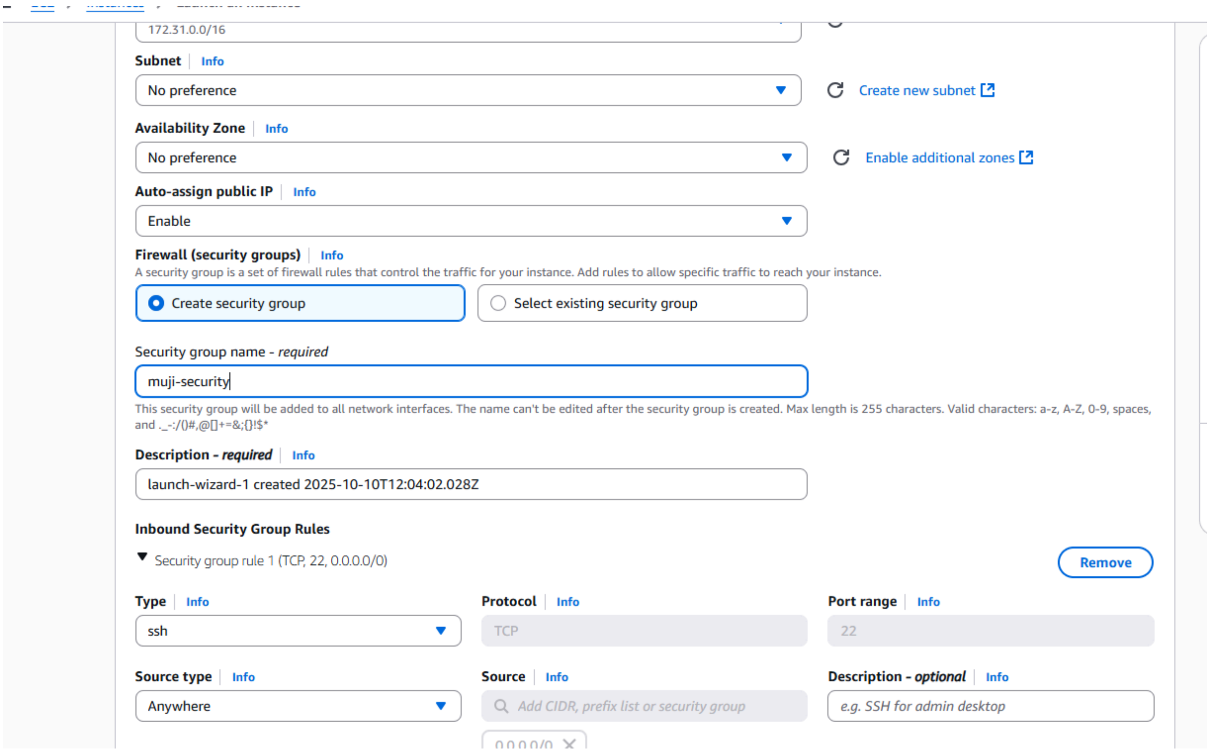




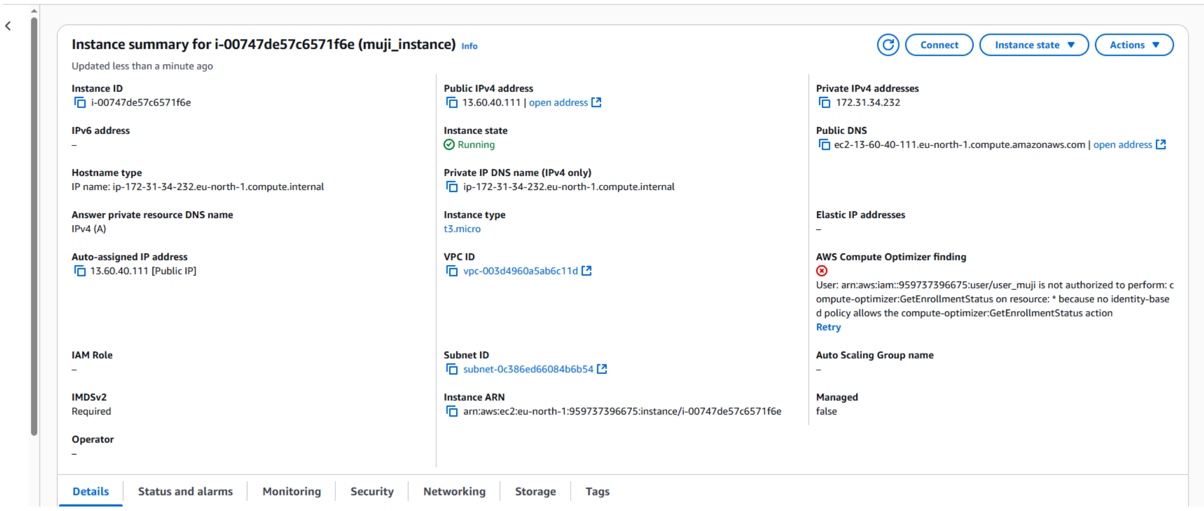
Create key


Create key pair
Key pair name
Key pairs allow you to connect to your instance securely.
my-for-server001
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more
Cancel
Create key pair


EC2 > Instances > Launch an instance
Network settings
Edit
Network
vpc-003d4960a5ab6c11d
Subnet
No preference (Default subnet in any availability zone)
Auto-assign public IP
Enable
Firewall (security groups)
Create security group
Select existing security group
We'll create a new security group called 'launch-wizard-1' with the following rules:
Allow SSH traffic from
Anywhere
0.0.0.0/0
Allow HTTPS traffic from the internet
Allow HTTP traffic from the internet
Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Configure storage
Advanced
gp3
Root volume, 3000 IOPS, Not encrypted
Add new volume
Click refresh to view backup information
0 x File systems
Advanced details
Summary
Number of instances
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code



Launch



Connect

Connect Info

Connect to an instance using the browser-based client.

EC2 Instance Connect | Session Manager | SSH client | EC2 serial console

Instance ID

i-00747de57c6571f6e (muji_instance)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is my-key-for-server001.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
   chmod 400 "my-key-for-server001.pem"
4. Connect to your instance using its Public DNS:
   ec2-13-60-40-111.eu-north-1.compute.amazonaws.com

Example:

ssh -i "my-key-for-server001.pem" ubuntu@ec2-13-60-40-111.eu-north-1.compute.amazonaws.com

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

ubuntu@ip-172-31-34-232: ~

```
 System information as of Fri Oct 10 12:21:33 UTC 2025

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   25.6% of 6.71GB   Processes:             112
  Memory usage: 24%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.34.232

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-34-232:~$
```

ubuntu@ip-172-31-34-232: ~

```
  Memory usage: 24%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.34.232

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applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-34-232:~$ hostname
ip-172-31-34-232
ubuntu@ip-172-31-34-232:~$
```

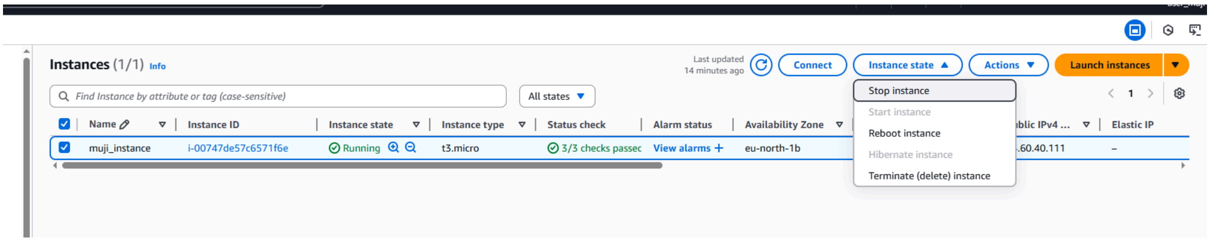
Instances (1/1) Info
Last updated
14 minutes ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Stop instance
Start instance
Reboot instance
Hibernate instance
Terminate (delete) instance
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Elastic IP
muji_instance
i-00747de57c6571f6e
Running
t3.micro
3/3 checks passec
View alarms
eu-north-1b
.60.40.111
–